

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_684a0618-ccb\agent-a050747f33fbd8a9d.jsonl`
- SHA-256: `eb026cc1ae3f63c21d00698f3fbc591df0c8943a6a977d6f5717ce3763e08480`
- Source modified: `2026-06-19T19:18:48+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_684a0618-ccb`
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T19:17:29.032Z)

THE IDEA TO ASSESS: a desktop app OR web UI where the user points KhelSutra at an ONLINE storage location they already own/pay for ? a Google Cloud Storage bucket, Google Drive, OneDrive/Microsoft 365, Dropbox, S3, etc. KhelSutra then WATCHES that folder, pulls/streams new recordings as they arrive, and AUTO-prepares highlights (same processing as the local/desktop path). "Bring your own storage + autosync": footage lands in their cloud ? reels appear automatically, zero manual upload.

You are assessing how this idea fits ONE area of the KhelSutra (badminton-highlight-indexer) docs. Read: docs/VIDEO_LOCALITY_MODEL.md. This is the "where the video lives" mental model. Does it already have a tier for "the user's own cloud storage"? Does it discuss autosync / watch-folder? Map the idea onto its tiers (e.g. L0 fully-local ? cloud).

Be evidence-based and honest ? quote real anchors, don't invent. If the doc does NOT anticipate the idea, say so plainly and note what it WOULD take. Return the structured object.

2. assistant (2026-06-19T19:17:34.015Z)

I'll read the target doc to assess how the BYO-storage/autosync idea fits.

3. user (2026-06-19T19:17:34.170Z)

```
1      # Video locality ? where the bytes live (mental model + open questions)
2
3      **Status:** THINKING DOC (owner-requested, 2026-06-12) ? a shared mental model for
4      the upload/storage problem, with the open questions named. **No implementation is
5      greenlit by this doc** (per CLAUDE.md, both tracks stay owner-gated).
6      **Companions:** `UI_PROPOSAL.md` (B2/B3 upload/download), `HOSTING_PLAN.md`,
7      `PLATFORM_ARCHITECTURE.md` §3 (future StorageBackend/ComputeBackend registries),
8      collector ADR-002 (the frame-identical proxy contract this doc leans on).
9
10     ## 0. The questions that triggered this (and the short answers)
11
12     ***"Do we store the video they upload?"** Yes ? today the chunked-upload flow
13     assembles the full file under `security.upload_dir` (default `output/uploads`) on
14     our box, registers it by MD5, and nothing ever deletes it (retention policy is an
15     acknowledged gap ? BACKLOG). Processing also strews derived artifacts (chunks,
16     frames, trajectories, WSL temp) and, on the Gemini path, uploads ?1280px
17     re-encoded chunks to Google's Files API.
18
19     ***"How would they upload a 10 GB file?"** Today they can't: `/api/upload/init`
20     rejects anything over `security.max_upload_mb` (default **4096 MB**), and even
21     under the cap the file moves as sequential ?90 MB parts through a free-tier edge
22     whose body limit forced that part size. For scale: 10 GB ? 30?45 min of 4K phone
23     video (or ~2?3 h at 1080p); on a typical Indian uplink that's **~1 h at 20 Mbps,
```

24 ~4.5 h at 5 Mbps** ? before retries. Browser-tab uploads of that size are
 25 miserable everywhere; expecting academies to manage S3/GCS buckets is a
 26 non-starter. **The 10 GB problem is real and the current design doesn't solve it.**
 27
 28 ***"Can we make it so there is NO upload?"** Mostly ? because of \$1.
 29
 30 ## 1. The central insight: timestamps are the product; the video is dead weight
 31
 32 Walk the pipeline and ask what each stage actually consumes:
 33
 34 - The **Gemini segmenter** never sees the original ? it re-encodes chunks down to
 35 **1280 px** (`indexing.ai\_chunk\_scale\_width`) before uploading them to Google.
 36 - The **detectors/featurizer** consume downsampled frames and trajectories.
 37 - The only consumer of the original full-res bytes is the **highlight compiler**,
 38 and only to cut clips ? a pure-ffmpeg operation that can run **wherever the
 39 original lives**.
 40 - What the customer receives is `{rally intervals, report, reel}`. Intervals +
 41 report are **kilobytes**. Given the intervals, the reel can be cut on the
 42 machine that already has the original.
 43
 44 So: **any design that moves the 10 GB is moving the wrong thing.** The question
 45 is never "how do we upload the video" but "what is the smallest artifact that
 46 must cross the network for this tier of service."
 47
 48 ## 2. The locality ladder
 49
 50 From most to least bytes leaving the academy's Windows desktop. Sizes are rough
 51 ratios for a 4K source ? measure on a real file before committing to any tier.
 52
 53 | | What crosses the network | ~Bytes (10 GB source) | What it needs | Status |
 54 |---|---|---|---|---|
 55 | **L3 ? full upload** (today's flow) | everything | 10 GB (blocked: 4 GB cap) | paid
 edge tier + resumable uploads + server disk + retention policy | built, capped, wrong for big
 files |
 56 | **L2 ? proxy upload** | a timeline-identical ?1080p re-encode | ~1?2.5 GB | a tiny
 client prep step (ffmpeg) ? **the exact frame-identical contract we already built and verified
 as collector `prep-annotation-proxy`** (same fps, same frame count, parity-checked, MD5?MD5
 manifest) | contract proven; client wrapper + server acceptance = new work |
 57 | **L1 ? chunks/candidates upload** | only the ?1280px AI chunks (or just candidate
 windows) the Gemini path needs anyway | ~0.3?1 GB | client-side chunk+downscale (ffmpeg, CPU);
 server becomes a **GPU-less orchestrator** | the server half exists (it's the gemini segmenter's
 own flow, relocated) |
 58 | **L1-direct** | chunks go **straight from the client to Gemini** ? zero video bytes
 through our server | ~0 through us | per-user key custody / metering / cost ceilings (OQ4) |
 idea only |
 59 | **L0 ? fully local** | nothing (calls to Google only if the Gemini path is used) | 0 |
 the owned on-device model (roadmap north star) or a local GPU + our stack | not alpha-feasible;
 the long-term answer |
 60 | **L4 ? sneakernet** | a hard drive | 0 over network | a person (the field associate /
 owner) | how the pilot actually works today |
 61
 62 **The honest "no upload" answer:** *strictly* no bytes leave only at L0 with the
 63 owned model (and even L0 leaks ?1280px chunks to Google while the Gemini path is
 64 the brain ? say that plainly in consent language, OQ3). The pragmatic near-term
 65 answer is **L2 or L1: don't eliminate the upload, shrink it 5?30x** ? at which
 66 point the existing 90 MB-part flow, the 4 GB cap, and a coffee-break wait are all
 67 suddenly fine.
 68
 69 ## 3. Second-order effects worth noticing
 70
 71 - **L1/L2 make the serving box GPU-less for the Gemini tier** ? ffmpeg + API
 72 orchestration only. Hosting cost collapses; the GPU box stays for WASB/owned-
 73 model work. (PLATFORM \$3's ComputeBackend split falls out naturally.)
 74 - **Storing the proxy instead of the original** is ~10x less disk AND a smaller
 75 privacy surface; the review UI (alpha screen A4) actually *prefers* streaming a

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76     1080p proxy to seeking inside a 10 GB original.
77 - **The correction loop survives** at every tier ?L1: labels are intervals on a
78   timeline, and the proxy timeline IS the original timeline by construction
79   (the collector ADR-002 invariants: same fps, same frame count, parity-verified).
80 - **video_id keying:** the indexer keys by MD5 of the file it processed. At L2
81   that's the proxy's MD5; the proxy-manifest pattern (original_md5 ? proxy_md5)
82   already exists ? but the product needs ONE registered identity rule (OQ2).
83
84 ## 4. The frugality frame ? afford the bills by not burning GPU
85
86 The runway constraint (owner, 2026-06-12): **the bills stay personally affordable
87 for a long time as long as we don't burn too much GPU.** The locality ladder maps
88 directly onto that:
89
90 - **The Gemini serving path costs ~$0.30?0.60 per hour of video** in API calls
91   (2026-06-10 quality table) and at L1/L2 needs **no GPU at all** ? a CPU box,
92   ffmpeg, and an API key. At pilot scale (a handful of academies, a few sessions
93   a week) that is single-digit dollars a month: affordable indefinitely.
94 - **GPU spend is the thing to guard.** WASB / owned-model inference and training
95   stay on the owner's already-paid-for local 2070S; rent nothing until the owned
96   model has a ship-gate-passing reason to be served. L2/L1 turn this from a
97   discipline into a default: *the serving box has no GPU to burn.*
98 - **Storage follows the same logic:** keep proxies, not originals (~10x less
99   disk); settle retention (OQ1) before volume arrives; the free edge tier stays
100  adequate precisely because L2 keeps payloads under the existing caps.
101 - **Put an explicit cost ceiling per video** (ties into OQ4): a 7-hour tape or a
102   runaway retry must not surprise-bill. The existing ops learnings (serial
103   uploads, `ai_max_workers=1`, bounded exponential retry) already point this way.
104
105 ## 5. The academy-visit dividend ? field work builds the product, not just the verdict
106
107 The C13 visits aren't only a market read; run deliberately, each visit deposits
108 product assets:
109
110 - **Phase-2 recordings (consented, adults-only) ARE the target corpus.** The
111   owned model's ship target (0.66-to-beat) lives on exactly the messy multi-court
112   footage class these academies produce ? every show-back visit grows the
113   training set the model is starved for, at zero marginal engineering cost.
114 - **The show-back is a live dress rehearsal of the product loop:** footage comes
115   back on a drive (L4), we proxy + process + return a reel ? the same motion
116   L2 will automate. Every friction the associate hits there (file handoff,
117   proxy time, reel delivery) is a pre-paid product-backlog item; write it down.
118 - **Demo reactions and improvement asks** (captured verbatim on the answer
119   sheet) are the build backlog, collected by someone already standing in front
120   of the user.
121 - **Connectivity notes from visits** answer OQ8 with data instead of guesses.
122
123 ## 6. Open questions (the decision backlog this doc exists to hold)
124
125 | # | Question | Why it matters / current lean |
126 |---|---|---|
127 | OQ1 | **Retention policy per artifact class** ? original (only L3 has it), proxy,
chunks, frames/trajectories, Gemini remote files, intervals/labels | intervals/labels are the
asset (keep forever, tiny); originals/proxies need an explicit keep-while-active + delete-on-
request rule; Gemini remote-file GC already in BACKLOG |
128 | OQ2 | **Which file's MD5 is the video's identity** at L2/L1 | lean: the processed
file's MD5 (proxy), with original_md5 carried in metadata ? the collector manifest pattern,
promoted to a product rule |
129 | OQ3 | **Consent language for the Gemini path** ? even "local" modes ship 1280px chunks
to Google | must be plain in the academy-facing terms; paid-tier no-training posture to be cited
|
130 | OQ4 | **Key custody for L1-direct** ? client calls Gemini with whose key? per-user
metering, abuse, cost ceiling per video | unsolved; this is what separates L1 from L1-direct |
131 | OQ5 | **Windows client packaging** ? portable ffmpeg + a signed tiny app vs a
PowerShell script for the pilot | pilot can be a script the associate/owner runs; product needs

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an installer story (ffmpeg LGPL build = licensing-clean) |
132 | OQ6 | **What does the review UI play** at each tier | L2 streams the proxy (good); L1
has only chunks ? review degrades ? this quietly argues L2 over L1 as the default |
133 | OQ7 | **Reel delivery** ? server-compiled from proxy (shareable 1080p) vs intervals-
back + client cuts full-res | could be both: proxy reel by default, "premium full-res" cut
locally |
134 | OQ8 | **Real academy uplink numbers** ? the L2-vs-L3 case rests on bandwidth
assumptions | C13 visits can note connectivity informally; don't redesign around guesses |
135 | OQ9 | **Recurring volume** ? an academy recording daily generates 10 GB/day; even L2
accumulates | retention (OQ1) + maybe future delta thinking; not an alpha problem |
136 | OQ10 | **If L3 must survive** (users who refuse any client-side step) | needs
resumable uploads + a paid edge tier + a raised cap; treat as the exception path, not the design
center |
137
138 ## 7. Where this lands (recommendation, not a greenlight)
139
140 **Default the alpha to L2 (proxy upload):** a one-command client prep step built
141 on the *already-verified* proxy contract, uploads that fit comfortably inside
142 today's chunked endpoint and 4 GB cap, a GPU-less serving path (the affordability
143 requirement, $4, satisfied by construction), a review UI that streams the proxy,
144 and intervals that map 1:1 onto the customer's original.
145 **L1/L1-direct is the fast-follow** once OQ4 has an answer; **L0 is the owned-
146 model milestone**, not an alpha decision. The field pilot keeps running on L4 ?
147 a person with a hard drive is a fine storage backend for n=3 academies.
148

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4. assistant (2026-06-19T19:17:38.328Z)

I have read the doc. It's a focused thinking doc on video locality. Let me check whether the companion sections it references add any BYO-cloud-storage tier, since the doc explicitly points to `PLATFORM\_ARCHITECTURE.md` §3 (StorageBackend registry) as the home for storage backends.

5. user (2026-06-19T19:17:40.468Z)

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15:browsable by history**, and choose **where their data lives** (Google Drive / cloud
16:bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
41:> Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
58:| Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
85: PLANE ? (local | BYO-VM | ? ? (per-tenant DB: ?
92: ? Storage (local upload | Google Drive | S3 | ?
93: ? GCS | Azure) ? raw video + highlight outputs ?
109:### 3.1 `StorageBackend` registry
114:class StorageBackend(ABC):
125: local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
126: Dropbox via `dropboxdrivefs`, reaching for **`smart_open`** as the robust MP4
127: byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
128: add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
129:- **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**:
130: the tenant registers their own bucket/Drive; raw video **never leaves their account**.
131:- **Auth ? never persist users' long-lived keys.** For buckets, have the user create an
132: IAM role you **`AssumeRole`** into, scoped to one bucket/prefix, and mint **presigned
133: URLs** so raw video moves **browser ? the user's bucket directly, never through our
134: servers**. For Drive, OAuth with the narrow **`drive.file`** scope; for academies, a
148:- **`byo_worker`** ? the tenant runs a **pull-based worker** (a small agent) on their own
155: credential-safe v1** for BYO-compute.
158: (BYOC: everything launches in the user's accounts/VPCs; creds stay with them).
165: treat `byo_worker`/SkyPilot as a **power-user / enterprise / privacy-sensitive tier**,
166: not the academy default. If/when BYO ships, plan for **auto-updating, self-healing
168:- **Middle tier to consider ? "zero-infra BYO":** a managed-but-isolated path where the
169: **storage** stays in the user's bucket (signed URLs) while compute runs on **our** GPUs
171: This gives the BYO *privacy* benefit (raw video never persists on our infra) **without**
172: the user running a worker ? a pragmatic default between full-hosted and full-BYO.
234:- **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs
235: for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,

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309:inference jobs (even ****BYO-compute**** can serve inference on the tenant's GPU); the
356:| ****ms-swift**** (ModelScope) | Self-managed (rented/own GPU) | ****Native video+audio****;
Qwen3-VL/Omni, 300+ MLLMs | any incl ****7B**** | ? DPO/GRPO | ? full ownership | ****? video-QA**;
slots into BYO-compute (§3.2)****** |
372: ****ComputeBackend** / BYO-compute** seam (§3.2). ****Likely a better fit than Tinker**** for
401:****Engine** (don't reinvent):**** ms-swift**** already delivers this for VLMs ? CLI-driven,
443: ****explicitly-contributed, licensed**** corpus ? ****not**** on whatever sits in a tenant's bucket
458: your storage, your compute ? and now ****your model****. With BYO-compute the adapter can
be
477:| ****P2**** | ****StorageBackend` registry + BYO-storage**** | new ABC+registry;
`local`:`gdrive`/`s3`/`gcs`; replace `video\_path` with `StorageRef`; encrypted per-tenant creds
|
478:| ****P3**** | ****ComputeBackend` registry + BYO-compute\*\*** | `local`/`hosted`/`byo_worker`;
pull-based worker agent; control/data-plane split |
479:| ****P4**** | ****Rich rally/shot semantic metadata & tagging**** *(the next value stage after
solid rally detection + highlights)* | shot type (smash / backhand drive / net), rally
****outcome**** (winner / forced / ****unforced error****), context (e.g. ****open court but error****) ?
query-API filters + `HighlightMetadata` + a ****human correction/review loop**** feeding the golden
set via the `annotation` registry + `candidate\_segments` |
486:that ended in a smash or backhand drive,"* "rallies where the court was open but the
507:- ****BYO-storage is the central privacy/liability lever:**** if raw video stays in the
508: tenant's own bucket/Drive and we process via signed URL or their compute, we minimize
518: what merely passes through a tenant's bucket. ****Exclude minors' video from the training**
532:1. ****Hosted-first vs BYO-first.**** Recommended: ****hosted control-plane first**** (P0?P1) with
533: ****BYO-storage soon after**** (P2); ****BYO-compute later**** (P3) as a power-user/enterprise
542:6. ****How far to push BYO-compute**** given its security/ops complexity.
552:tests); BYO-compute security; ****biometric/minor liability**** (get this right before any
564:[Omitted long matching line]
568:[Omitted long matching line]

6. assistant (2026-06-19T19:17:45.811Z)

The companion doc `PLATFORM\_ARCHITECTURE.md` has extensive BYO-storage design (a
`StorageBackend` registry with `gdrive`/`s3`/`gcs`/`azure`/`dropbox` backends). But the question
is specifically about `VIDEO\_LOCALITY\_MODEL.md`. Let me verify whether the locality doc mentions
a watch-folder/autosync notion at all, and check the §2 ladder precisely for whether "user's own
cloud" is a distinct tier vs. a passing reference.

The doc is short and I've read it fully. Let me confirm there's no
"watch"/"poll"/"sync"/"automatic"/"trigger on arrival" language in it.

7. user (2026-06-19T19:17:51.438Z)

7:`PLATFORM\_ARCHITECTURE.md` §3 (future StorageBackend/ComputeBackend registries),
10:## 0. The questions that triggered this (and the short answers)
25:miserable everywhere; expecting academies to manage S3/GCS buckets is a
60:| ****L4 ? sneakernet**** | a hard drive | 0 over network | a person (the field associate /
owner) | how the pilot actually works today |
99: disk); settle retention (OQ1) before volume arrives; the free edge tier stays
115: back on a drive (L4), we proxy + process + return a reel ? the same motion
116: L2 will automate. Every friction the associate hits there (file handoff,
135:| OQ9 | ****Recurring volume**** ? an academy recording daily generates 10 GB/day; even L2
accumulates | retention (OQ1) + maybe future delta thinking; not an alpha problem |
147:a person with a hard drive is a fine storage backend for n=3 academies.

8. assistant (2026-06-19T19:17:56.116Z)

I now have a complete, evidence-based picture. The `VIDEO\_LOCALITY\_MODEL.md` doc does NOT have a
tier for "the user's own cloud storage" and never discusses autosync/watch-folder. Its entire
ladder is framed around *what artifact crosses the network from the academy desktop to OUR box*,
not around the video already living in the user's cloud. The only mention of S3/GCS buckets
(line 25) is dismissive ("expecting academies to manage S3/GCS buckets is a non-starter"). The
BYO-storage concept lives entirely in the companion `PLATFORM\_ARCHITECTURE.md` §3, which this
doc only points to.

9. user (2026-06-19T19:18:39.285Z)

Structured output provided successfully

10. assistant (2026-06-19T19:18:48.533Z)

I've assessed the idea against `docs/VIDEO\_LOCALITY\_MODEL.md` and returned the structured object.

Bottom line: the locality doc does NOT anticipate BYO-cloud-storage + autosync. Its entire L0?L4 ladder classifies designs by "smallest artifact that must cross OUR network *from the academy desktop*", not by "the original lives in the user's own cloud". The words watch/poll/sync/webhook never appear; the only S3/GCS mention (line 25) is dismissive. The actual home for this idea is the companion `docs/PLATFORM\_ARCHITECTURE.md` §3.1 `StorageBackend` registry (which already designs `gdrive`/`s3`/`gcs`/`azure` backends, OAuth `drive.file` scopes, and presigned-URL custody) plus a new watch/sync trigger feeding the committed-next async job model. Conceptually it maps best as an "automated L4" ? the networked, unattended cousin of sneakernet ? and the doc's own §1 insight ("cut the reel wherever the original lives") is precisely what makes it viable, but that connection is not drawn here.